



Prototype of Cruz's metal 3D printer

an additive manufacturing process that reduces the amount of wastage of materials. The printer works by depositing powdered metals and sintering them to form a 3D object. With R&D for more than five years, Agustin has developed a 3D printer that uses an electron beam for sintering the metal powder.

"The most important technical challenges are already 70% solved. The functional prototype is composed of a vacuum chamber made of stainless steel, a vacuum pump system, an electron gun, high voltage power sources, electrical connections, cooling systems, sensors, among others," says Cruz. "This printer uses an electron beam to selectively melt each layer of metal powder, causing the powder particles to fuse. After one layer is completed, the build platform is moved down one layer in height. The re-coater comes in again with a fresh layer of powder, and the electron beam starts to induce the fusion of powder particles, causing the new layer to form. This process is repeated until the entire part is finished, making a solid metal part."

Cruz believes that to keep the metal printer accessible for all, it is necessary to utilise open source hardware and software. "This 3D metal printer will use an already

available open source 3D printing software—like Cura, Slic3r, and others—and the hardware algorithm is programmed in Arduino boards to make it easier to modify and distribute by anyone," he says.

Agustin expects that his tool will not only help innovators but people from all walks of life. It can be extremely useful in hospitals in remote locations. "3D metal printing will be practical when supplies run low in hospitals located away from major cities. For example, patients from the nearby fields who come in with work-related injuries could be outfitted with custom splints that are custom printed as needed," says Cruz. "We also find that entrepreneurs will benefit from this project, especially early-stage entrepreneurs that are not yet financially secure enough to easily afford a consumer 3D metal printer. Increasing access for these entrepreneurs ensures greater representation and impact, paving pathways to get these deep-tech innovations more efficiently into the market."

Currently, Cruz is working full-time in the development of this metal 3D printer, which, according to him, is an extremely resource-intensive job. Hence, he plans to open the project to the community and jump on the crowdfunding bandwagon. **EFY**

—Sharad Bhowmick



EFY GROUP
YOURS SINCE 1969

Facing problems in getting EFY Magazine in your area?

If so, let us know!



Electronics For You magazine is available at all top news-stands/outlets in India. But if you are facing problem in getting EFY magazine in your area, please call Akhilesh Yadav on **+91-8800295612** or drop an email at efycirc@efy.in. We will help you locate the nearest news-stand or arrange a copy at your doorstep.

electronics
FOR YOU

South Asia's Most Popular Electronics Magazine

Those who wish to distribute EFY Group publications may also call Akhilesh Yadav on +91-8800295612 or write at efycirc@efy.in